

CHAPTER 6

Stickelberger's Theorem

The aim of this chapter is to give, for any abelian number field, elements of the group ring of the Galois group which annihilate the ideal class group. They will form the Stickelberger ideal. The proof involves factoring Gauss sums as products of prime ideals, and since Gauss sums generate principal ideals, we obtain relations in the ideal class group. As an application, we prove Herbrand's theorem which relates the nontriviality of certain parts of the ideal class group of $\mathbb{Q}(\zeta_p)$ to p dividing corresponding Bernoulli numbers. Then we calculate the index of the Stickelberger ideal in the group ring for $\mathbb{Q}(\zeta_{p^n})$ and find it equals the relative class number. Finally, we prove a result, essentially due to Eichler, on the first case of Fermat's Last Theorem. In the next chapter we shall use Stickelberger elements to give Iwasawa's construction of p -adic L -functions.

§6.1. Gauss Sums

In order to prove Stickelberger's theorem, we need to study Gauss sums, which are also interesting in their own right. The Gauss sums used here are not the same as those used earlier, but there are many similarities.

Let $\mathbb{F} = \mathbb{F}_q$ be the finite field with q elements, q being a power of the prime p . Let ζ_p be a fixed primitive p th root of unity and let T be the trace from $\mathbb{F}$ to $\mathbb{Z}/p\mathbb{Z}$. Define

$$\psi: \mathbb{F} \rightarrow \mathbb{C}^\times, \quad \psi(x) = \zeta_p^{T(x)},$$

which is easily seen to be a well-defined, nontrivial (T is surjective) character of the additive group of $\mathbb{F}$. Let

$$\chi: \mathbb{F}^\times \rightarrow \mathbb{C}^\times$$

be a multiplicative character of $\mathbb{F}^\times$. We extend χ to all of $\mathbb{F}$ by setting $\chi(0) = 0$ (even if χ is the trivial character). Note that χ^{q-1} is the trivial character, so the order of χ is prime to p . If $q \neq p$, such characters are not the Dirichlet characters studied earlier. The concept of conductor will not enter into the present discussion.

Define the Gauss sum

$$g(\chi) = - \sum_{a \in \mathbb{F}} \chi(a) \psi(a)$$

If χ has order m , then $g(\chi) \in \mathbb{Q}(\zeta_{mp})$. A quick calculation shows that $g(1) = 1$.

Lemma 6.1. (a) $g(\bar{\chi}) = \chi(-1)\overline{g(\chi)}$;
 (b) if $\chi \neq 1$, $g(\chi)g(\bar{\chi}) = \chi(-1)q$;
 (c) if $\chi \neq 1$, $g(\chi)g(\chi) = q$.

Proof. (a) is straightforward. (b) follows from (a) and (c). For (c),

$$\begin{aligned} g(\chi)g(\bar{\chi}) &= \sum_{a, b \neq 0} \chi(ab^{-1})\psi(a-b) \\ &= \sum_{b, c \neq 0} \chi(c)\psi(bc-b) \quad (\text{let } c = ab^{-1}) \\ &= \sum_{b \neq 0} \chi(1)\psi(0) + \sum_{c \neq 0, 1} \chi(c) \sum_{b \neq 0} \psi(b(c-1)) \\ &= (q-1) + \sum_{c \neq 0, 1} \chi(c)(-1) = q. \end{aligned}$$

This completes the proof. □

If χ_1, χ_2 are two multiplicative characters, we define the Jacobi sum

$$J(\chi_1, \chi_2) = - \sum_{a \in \mathbb{F}} \chi_1(a)\chi_2(1-a).$$

More generally, we have

$$J(\chi_1, \dots, \chi_n) = (-1)^{n-1} \sum_{a_1 + \dots + a_n = 1} \chi_1(a_1) \dots \chi_n(a_n),$$

but we do not need this for $n > 2$. Note that if χ_1 and χ_2 have orders dividing m then $J(\chi_1, \chi_2)$ is an algebraic integer in $\mathbb{Q}(\zeta_m)$.

Lemma 6.2. (a) $J(1, 1) = 2 - q$;
 (b) $J(1, \chi) = J(\chi, 1) = 1$ if $\chi \neq 1$;
 (c) $J(\chi, \bar{\chi}) = \chi(-1)$ if $\chi \neq 1$;
 (d) $J(\chi_1, \chi_2) = g(\chi_1)g(\chi_2)/g(\chi_1\chi_2)$ if $\chi_1\chi_2 \neq 1$.

Proof. (a) and (b) are easy. To prove (c) and (d), we compute

$$\begin{aligned}
g(\chi_1)g(\chi_2) &= \sum_{a,b} \chi_1(a)\chi_2(b)\psi(a+b) \\
&= \sum_{a,b} \chi_1(a)\chi_2(b-a)\psi(b) \\
&= \sum_{\substack{a,b \\ b \neq 0}} \chi_1(a)\chi_2(b-a)\psi(b) + \sum_a \chi_1(a)\chi_2(-a).
\end{aligned}$$

If $\chi_1\chi_2 \neq 1$, then the second sum vanishes. If $\chi_1\chi_2 = 1$, then it equals $\chi_1(-1)(q-1)$. The first sum equals (let $a = bc$)

$$\sum_{\substack{b,c \\ b \neq 0}} \chi_1(b)\chi_2(b)\chi_1(c)\chi_2(1-c)\psi(b) = g(\chi_1\chi_2)J(\chi_1, \chi_2).$$

If $\chi_1\chi_2 \neq 1$, we obtain (d). If $\chi_1\chi_2 = 1$, use Lemma 6.1(b), along with $g(1) = 1$, to obtain (c). This completes the proof. $\square$

Corollary 6.3. *If χ_1, χ_2 are characters of orders dividing m , then*

$$\frac{g(\chi_1)g(\chi_2)}{g(\chi_1\chi_2)}$$

is an algebraic integer in $\mathbb{Q}(\zeta_m)$.

Proof. If $\chi_1\chi_2$ is nontrivial, use the above result. The remaining cases are quickly checked individually. $\square$

The significance of this result is twofold: not only is the expression integral, is also eliminates ζ_p . This will be useful later.

Let m be an integer with $(m, p) = 1$. Then the fields $\mathbb{Q}(\zeta_m)$ and $\mathbb{Q}(\zeta_p)$ are disjoint. Let $(b, m) = 1$. We may define $\sigma_b \in \text{Gal}(\mathbb{Q}(\zeta_m, \zeta_p)/\mathbb{Q})$ by

$$\sigma_b: \zeta_p \mapsto \zeta_p, \quad \zeta_m \mapsto \zeta_m^b.$$

(perhaps it would be better to use double indices and call this $\sigma_{1,b}$, but usually ζ_p will drop out early, leaving only ζ_m).

Lemma 6.4. *Assume χ^m is trivial. Then*

$$\frac{g(\chi)^b}{g(\chi)^{\sigma_b}} = g(\chi)^{b-\sigma_b} \in \mathbb{Q}(\zeta_m),$$

and $g(\chi)^m \in \mathbb{Q}(\zeta_m)$.

Proof. The second follows from the first if we let $b = 1 + m$. For the first, we have

$$g(\chi)^{\sigma_b} = -\sum \chi(a)^b \psi(a) = g(\chi^b).$$

Let $\tau \in \text{Gal}(\mathbb{Q}(\zeta_{mp})/\mathbb{Q}(\zeta_m))$, so $\tau: \zeta_m \mapsto \zeta_m, \zeta_p \mapsto \zeta_p^c$ for some $c, (c, p) = 1$. Then

$$\begin{aligned}
g(\chi)^\tau &= -\sum \chi(a)\psi(ca) \\
&= -\chi(c)^{-1} \sum \chi(a)\psi(a) = \chi(c)^{-1}g(\chi),
\end{aligned}$$

and similarly

$$g(\chi^b)^\tau = \chi(c)^{-b} g(\chi^b).$$

Therefore τ fixes $g(\chi)^{b^{-\sigma_b}}$. The result follows. $\square$

Lemma 6.5. $g(\chi^p) = g(\chi)$.

Proof. Since $a \mapsto a^p$ is an automorphism over $\mathbb{Z}/p\mathbb{Z}$, $T(a) = T(a^p)$, and a^p yields a permutation of $\mathbb{F}$. Therefore

$$\begin{aligned} g(\chi^p) &= -\sum \chi(a^p) \zeta_p^{T(a)} \\ &= -\sum \chi(a^p) \zeta_p^{T(a^p)} = g(\chi). \end{aligned} \quad \square$$

This completes our list of basic properties of Gauss sums. We now digress to give an application to the Fermat curve.

We wish to count the number of solutions of

$$X^d + Y^d = 1, \quad \text{with } X, Y \in \mathbb{F}_q.$$

As is usually the case, it is more natural to count points in projective space. That is, we consider solutions, except $(0, 0, 0)$, of

$$X^d + Y^d = Z^d,$$

and identify two solutions if they differ by a scalar multiple. If $Z \neq 0$, we may identify (X, Y, Z) with $(X/Z, Y/Z, 1)$ and obtain a solution of the original equation. But if $Z = 0$, we obtain the “points at infinity” ($X/0 = \infty$), which correspond to solutions of $X^d + Y^d = 0$. Since we do not count $(0, 0, 0)$, we must have $Y \neq 0$, so any solution $(X, Y, 0)$ may be put in the form $(X/Y, 1, 0)$. The number of points at infinity is exactly the number of solutions in $\mathbb{F}_q$ of $X^d = -1$.

Despite all this, we shall start by counting the solutions of $X^d + Y^d = 1$, and make the correction later. We first assume d divides $q - 1$. Since $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, there exists a character χ of $\mathbb{F}_q^\times$ of order exactly d . The cyclicity implies that $\chi(u) = 1$ if and only if $u \in \mathbb{F}_q^\times$ is a d th power. For $u \in \mathbb{F}_q^\times$, let $N_d(u)$ be the number of solutions in $\mathbb{F}_q$ of $X^d = u$, so

$$N_d(u) = \begin{cases} 1, & u = 0 \\ 0, & u \neq 0, u \neq d\text{th power} \\ d, & u \neq 0, u = d\text{th power (since } d|q-1). \end{cases}$$

It follows easily that

$$N_d(u) = \sum_{a=1}^d \chi^a(u) \quad \text{if } u \neq 0.$$

Therefore, the number of solutions of $X^d + Y^d = 1$ is

$$\sum_{\substack{u+v=1 \\ uv \neq 0}} N_d(u) N_d(v) + 2d$$

(the second term corresponds to $X = 0$ or $Y = 0$)

$$\begin{aligned} &= 2d + \sum_{u \neq 0, 1} \sum_{a=1}^d \sum_{b=1}^d \chi^a(u) \chi^b(1-u) \\ &= 2d - \sum_{a=1}^d \sum_{b=1}^d J(\chi^a, \chi^b). \end{aligned}$$

From Lemma 6.2 we see that the term $a = b = d$ contributes $2 - q$; the terms with either $a = d$ or $b = d$, but not both, contribute a total of $2(d - 1)$; those with $a + b = d$ yield $\sum_{a=1}^{d-1} \chi^a(-1) = N_d(-1) - 1$; and the remaining terms can be expressed in terms of Gauss sums via Lemma 6.2(d). Using the fact that $N_d(-1)$ is the number of points at infinity, we find that the number of solutions of $X^d + Y^d = Z^d$ in projective space is

$$q + 1 - \sum_{\substack{a, b=1 \\ a+b \neq d}}^{d-1} J(\chi^a, \chi^b).$$

Since $|g(\chi)| = \sqrt{q}$ if $\chi \neq 1$, we have, using Lemma 6.2(d), that

$$|\sum J(\chi^a, \chi^b)| \leq (d - 1)(d - 2)\sqrt{q}.$$

If $\bar{N}$ denotes the number of solutions,

$$|\bar{N} - (q + 1)| \leq (d - 1)(d - 2)\sqrt{q}.$$

This is a special case of a more general result which states that for a curve of genus g we have

$$|\bar{N} - (q + 1)| \leq 2g\sqrt{q}.$$

Note that $q + 1$ is the number of points on a line $aX + bY = cZ$, so the number of points on a curve is approximately the same as for a line, the possible error being bounded in terms of the genus.

Now assume d is arbitrary, so we do not necessarily have $d|q - 1$. Let $e = (d, q - 1)$. Then $N_d(u) = N_e(u)$, so

$$\begin{aligned} \bar{N} &= \# \{X^d + Y^d = Z^d | (X, Y, Z) \neq (0, 0, 0)\} / (q - 1) \\ &= \sum_{u+v=w} N_d(u) N_d(v) N_d(w) / (q - 1) \\ &= \# \{X^e + Y^e = Z^e | (X, Y, Z) \neq (0, 0, 0)\} / (q - 1). \end{aligned}$$

Since e divides $q - 1$, we have

$$|\bar{N} - (q + 1)| \leq (e - 1)(e - 2)\sqrt{q} \leq (d - 1)(d - 2)\sqrt{q},$$

so we have proved the following.

Proposition 6.6. *Let $\bar{N}$ denote the number of projective space solutions of*

$$X^d + Y^d = Z^d$$

in $\mathbb{F}_q$. Then

$$|\bar{N} - (q + 1)| \leq (d - 1)(d - 2)\sqrt{q}. \quad \square$$

Corollary 6.7. *For any given d , $X^d + Y^d \equiv 1 \pmod{p}$ has solutions with $XY \not\equiv 0 \pmod{p}$, for all sufficiently large p .*

Proof. From the above, the number of points at infinity is $N_d(-1)$ (or $N_e(-1)$), which is at most d . The number of solutions with $X \equiv 0$ or $Y \equiv 0$ is at most $2d$. Therefore we have a nontrivial solution as soon as $\bar{N} > 3d$. Since $\bar{N} - p = O(\sqrt{p})$, the result follows. $\square$

This corollary shows that it would be difficult to prove Fermat's Last Theorem using only congruences.

To finish this digression we show that Proposition 6.6 is essentially the Riemann hypothesis for the Fermat curve. Fix d and p and let $\bar{N}_n$ be the number of solutions of $X^d + Y^d = Z^d$ in projective space over $\mathbb{F}_{p^n}$. The zeta function $\zeta_d(s)$ of the curve may be defined as follows:

Define $Z(T)$ by

$$\frac{Z'(T)}{Z(T)} = \sum_{n=1}^{\infty} \bar{N}_n T^{n-1}, \quad Z(0) = 1.$$

Then

$$\zeta_d(s) = Z(p^{-s}).$$

This function satisfies many properties similar to those for Dedekind zeta functions: for example, there is an Euler product, and also there is a functional equation relating the values at s and $1 - s$. It can be shown that $Z(T)$ is a rational function of the form

$$Z(T) = \frac{P(T)}{(1 - T)(1 - pT)}, \quad \text{where } P(T) \in \mathbb{Z}[T], P(0) = 1$$

(for further properties and proofs, see Weil [6] or Eichler [3]).

Writing $P(T) = \prod_i (1 - \alpha_i T)$, we see that

$$\frac{Z'(T)}{Z(T)} = \sum_{n=1}^{\infty} \left(1 + p^n - \sum_i \alpha_i^n \right) T^{n-1}.$$

so

$$\bar{N}_n = 1 + p^n - \sum_i \alpha_i^n.$$

To answer the question that arises when one compares this with a previous formula, yes, the α_i 's are Jacobi sums, but we shall not prove this here. It follows easily from the Davenport–Hasse relations (see the Exercises).

From Proposition 6.6 we have

$$\left| \sum_i \alpha_i^n \right| \leq (d - 1)(d - 2)p^{n/2}.$$

Lemma 6.8.

$$\limsup_{n \rightarrow \infty} \left| \sum_i \alpha_i^n \right|^{1/n} = \max_i |\alpha_i|.$$

Proof. The only problem arises when two α 's have the same absolute value, in which case a straightforward proof would have to show that "cancellation" does not decrease the lim sup. However, there is the following classical trick. Consider the complex function

$$f(z) = \sum_i \frac{1}{1 - \alpha_i z} = \sum_{n=0}^{\infty} \left(\sum_i \alpha_i^n \right) z^n.$$

The radius of convergence of the power series is the distance to the nearest singularity, namely $1/\max |\alpha_i|$. But it is also the reciprocal of $\limsup |\sum \alpha_i^n|^{1/n}$. The result follows. $\square$

We now have $|\alpha_i| \leq \sqrt{p}$ for each i . Returning to $\zeta_d(s)$, we see that $\zeta_d(s) = 0 \Leftrightarrow p^s = \alpha_i$ for some i . Therefore $\operatorname{Re}(s) \leq \frac{1}{2}$. But the functional equation for $\zeta_d(s)$ implies that if $\zeta_d(s) = 0$ then $\zeta_d(1-s) = 0$. Therefore $\operatorname{Re}(s) = \frac{1}{2}$ for each zero s . This is the Riemann hypothesis for the Fermat curve.

All of the above is part of a much more general situation, which applies not only to curves but also to higher dimensional varieties (the Weil conjectures, now Deligne's theorem). The reader is strongly urged to read the classic papers of Weil ([1], [2]), where this is discussed and where additional results on Gauss and Jacobi sums are proved.

§6.2. Stickelberger's Theorem

Let $M/\mathbb{Q}$ be a finite abelian extension, so $M \subseteq \mathbb{Q}(\zeta_m)$ for some m (by the Kronecker–Weber theorem, proved in Chapter 14). We assume m is minimal. $G = \operatorname{Gal}(M/\mathbb{Q})$ may be regarded as a quotient of $(\mathbb{Z}/m\mathbb{Z})^\times$. We let σ_a , $(a, m) = 1$, denote both the element of $\operatorname{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ and its restriction to M . Let $\{x\}$ denote the fractional part of the real number x ; so $x - \{x\} \in \mathbb{Z}$ and $0 \leq \{x\} < 1$. Define the *Stickelberger element*

$$\theta = \theta(M) = \sum_{\substack{a \pmod{m} \\ (a, m) = 1}} \left\{ \frac{a}{m} \right\} \sigma_a^{-1} \in \mathbb{Q}[G].$$

The *Stickelberger ideal* $I(M)$ is defined to be $\mathbb{Z}[G] \cap \theta \mathbb{Z}[G]$, in other words, those $\mathbb{Z}[G]$ -multiples of θ which have integral coefficients.

Lemma 6.9. Suppose $M = \mathbb{Q}(\zeta_m)$. Let I' be the ideal of $\mathbb{Z}[G]$ generated by elements of the form $c - \sigma_c$, with $(c, m) = 1$. Let $\beta \in \mathbb{Z}[G]$. Then

$$\beta \theta \in \mathbb{Z}[G] \Leftrightarrow \beta \in I'.$$

Therefore $I = I'\theta$.

Proof. Since

$$(c - \sigma_c)\theta = \sum_a \left(c \left\{ \frac{a}{m} \right\} - \left\{ \frac{ac}{m} \right\} \right) \sigma_a^{-1} \in \mathbb{Z}[G],$$

we have “ $\Leftarrow$ ”. To prove the converse, first note that $m = (1 + m) - \sigma_{1+m} \in I'$. Suppose $(\sum_a x_a \sigma_a)\theta \in \mathbb{Z}[G]$, with $x_a \in \mathbb{Z}$. A short calculation shows that

$$\left(\sum_a x_a \sigma_a \right) \left(\sum_c \left\{ \frac{c}{m} \right\} \sigma_c^{-1} \right) = \sum_b \left(\sum_a x_a \left\{ \frac{ab}{m} \right\} \right) \sigma_b^{-1}.$$

Looking at the coefficient of σ_1^{-1} , we find that m divides $\sum x_a a$. Since $m \in I'$, so is $\sum x_a a$. Therefore

$$\sum x_a \sigma_a = \sum x_a (\sigma_a - a) + \sum x_a a \in I'.$$

This completes the proof. $\square$

This result is not true in general if M is a proper subfield of $\mathbb{Q}(\zeta_m)$. The problem is that $\sigma_b = 1$ for several b , so the “coefficient of σ_1^{-1} ” involves a sum over various values of b . For example, let $M = \mathbb{Q}(\sqrt{12}) = \mathbb{Q}(\zeta_{12} + \zeta_{12}^{11}) \subset \mathbb{Q}(\zeta_{12})$. Then $\sigma_1 = \sigma_{11} = 1$, while $\sigma_5 = \sigma_7 = \sigma$, say. We have $\theta(M) = 1 + \sigma \in \mathbb{Z}[G]$, so $1 \cdot \theta \in \mathbb{Z}[G]$. But I' is generated by $\{5 - \sigma, 7 - \sigma, 11 - 1\}$, therefore by $\{2, 1 + \sigma\}$. In particular, $1 \notin I'$.

If $x = \sum x_\sigma \sigma \in \mathbb{Z}[G]$ then x acts on ideals and ideal classes in the natural way: $A^x = \prod_\sigma (A^\sigma)^{x_\sigma}$.

Theorem 6.10 (Stickelberger's Theorem). *Let A be a fractional ideal of M , let $\beta \in \mathbb{Z}[G]$, and suppose $\beta\theta \in \mathbb{Z}[G]$. Then $A^{\beta\theta}$ is principal. Therefore, the Stickelberger ideal annihilates the ideal class group of M .*

Before starting the proof, we give two examples.

(a) Suppose M is real. Then $\sigma_a = \sigma_{-a}$ and $\{a/m\} + \{-a/m\} = 1$, so

$$\theta(M) = \frac{1}{2} \sum_{a \pmod{m}} \sigma_a = \frac{\phi(m)}{2 \deg M} \text{Norm}_{M/\mathbb{Q}}.$$

In this case we find that the norm, or some multiple of it, annihilates the ideal class group. This of course is already obvious, since $\mathbb{Q}$ has class number one. We therefore can obtain nontrivial results only if we look at imaginary fields.

(b) Suppose $M = \mathbb{Q}(\sqrt{-m})$ is imaginary quadratic. $\text{Gal}(M/\mathbb{Q}) = \{1, \sigma\}$, where σ is complex conjugation. Since $A^{1+\sigma}$ is an ideal of $\mathbb{Q}$, hence principal, σ acts by inversion on the ideal class group. Let $\beta = m$. Then $\beta\theta = \sum a \sigma_a^{-1} \in \mathbb{Z}[G]$, and in the ideal class group $\beta\theta$ acts as $\sum a \chi(a)$, where χ is the quadratic character for M . We find that $\sum a \chi(a) = m B_{1,\chi}$ annihilates the ideal class group. Of course, the class number formula implies that this number is just $-mh$ (if $m > 4$), so what we have is a weak

form of the analytic class number formula; however, it will be proved algebraically.

More generally, let F be any totally real number field and M/F a finite abelian extension. Let $G = \text{Gal}(M/F)$. Via the Artin map $A \mapsto \sigma_A \in G$, one can define partial zeta functions for $\sigma \in G$:

$$\zeta_F(\sigma, s) = \sum_{\sigma_A = \sigma} \frac{1}{N A^s} \quad (\text{Re}(s) > 1).$$

These may be meromorphically continued to the whole complex plane, and the values $\zeta_F(\sigma, -n)$ are rational numbers for $n \geq 0$. Define

$$\theta_n(M/F) = \sum_{\sigma \in G} \zeta_F(\sigma, -n) \sigma^{-1},$$

and let $I_n(M/F)$ be the ideal generated by elements of the form

$$(N A^{n+1} - \sigma_A) \theta_n(M/F),$$

where A ranges over a set of integral ideals of F not divisible by a certain finite set of prime ideals. Then $I_n(M/F)$ should annihilate some natural object, perhaps a K -group. For example, $I_1(M/\mathbb{Q})$ annihilates $K_2 \mathcal{O}_M$, except possibly for the 2-part, where $\mathcal{O}_M$ is the ring of integers of M . When $M = \mathbb{Q}(\zeta_m)$ and $F = \mathbb{Q}$, we have

$$\zeta(\sigma_a, s) = \sum_{\substack{b \equiv a(m) \\ b > 0}} \frac{1}{b^s},$$

which is essentially a Hurwitz zeta function. If $n = 0$, we have

$$\theta_0(M/\mathbb{Q}) = \sum_{c \pmod{m}} \left(\frac{1}{2} - \left\{ \frac{c}{m} \right\} \right) \sigma_c^{-1},$$

which differs from θ by half the norm (in fact, we shall need θ_0 later). Since $K_0 \mathcal{O}_M$ is essentially the ideal class group of M , the above may be regarded as an appropriate generalization of Stickelberger's Theorem. For details, see Coates [7].

We are now ready to start the proof of Stickelberger's theorem. The major step will be the factorization of certain Gauss sums. Let p be a prime and let $q = p^f$ be a power of p . Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Q}(\zeta_{q-1})$ lying above p . Since $\mathbb{Z}[\zeta_{q-1}] \pmod{\mathfrak{p}}$ is the finite field with q elements ($f = \text{residue class degree}$ by Theorem 2.13), and since the $(q-1)$ st roots of unity are distinct mod $\mathfrak{p}$, there is an isomorphism

$$\omega = \omega_{\mathfrak{p}}: \mathbb{F}_q^\times \rightarrow (q-1)\text{st roots of } 1$$

satisfying

$$\omega(a) \pmod{\mathfrak{p}} = a \in \mathbb{F}_q^\times.$$

This ω is essentially a generalization of the ω of the previous chapter. Let $\tilde{\mathcal{P}}$ be the prime of $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ lying above $\mathfrak{p}$. For $\alpha \in \mathbb{Z}$, let $s(\alpha) = v_{\tilde{\mathcal{P}}}(g(\omega^{-\alpha}))$, where $v_{\tilde{\mathcal{P}}}$ is the $\tilde{\mathcal{P}}$ -adic valuation. Clearly $s(\alpha)$ depends only on $\alpha \pmod{q-1}$.

- Lemma 6.11.** (a) $s(0) = 0$;
 (b) $0 \leq s(\alpha + \beta) \leq s(\alpha) + s(\beta)$;
 (c) $s(\alpha + \beta) \equiv s(\alpha) + s(\beta) \pmod{p-1}$;
 (d) $s(p\alpha) = s(\alpha)$;
 (e) $\sum_{\alpha=1}^{q-2} s(\alpha) = (q-2)(f)(p-1)/2$.

Proof. (a) is obvious; (b) and (d) follow from Corollary 6.3 and Lemma 6.5, respectively. Since $\tilde{\mathcal{P}}^{p-1} = \mathfrak{p}$, the values of $v_{\tilde{\mathcal{P}}}$ on $\mathbb{Q}(\zeta_{q-1})$ are divisible by $p-1$. Therefore (c) also follows from Corollary 6.3. Since $g(\omega^{-\alpha})g(\omega^{\alpha}) = \pm q = \pm p^f$, we have $s(\alpha) + s(q-1-\alpha) = v_{\tilde{\mathcal{P}}}(p^f) = (p-1)f$. Pairing up the terms in the sum (the term for $\alpha = (q-1)/2$ pairs with itself), we obtain (e). This completes the proof. $\square$

Lemma 6.12. $s(\alpha) > 0$ if $\alpha \not\equiv 0 \pmod{q-1}$, and $s(1) = 1$.

Proof. Since $\pi = \zeta_p - 1 \in \tilde{\mathcal{P}}$,

$$g(\omega^{-\alpha}) = -\sum \omega^{-\alpha}(a) \zeta_p^{T(a)} \equiv -\sum \omega^{-\alpha}(a) \equiv 0 \pmod{\tilde{\mathcal{P}}}.$$

Therefore $s(\alpha) > 0$. Also,

$$\begin{aligned} g(\omega^{-1}) &= -\sum \omega^{-1}(a) \zeta_p^{T(a)} \\ &= -\sum \omega^{-1}(a) (1 + \pi)^{T(a)} \equiv -\sum \omega^{-1}(a) (1 + \pi T(a)) \pmod{\tilde{\mathcal{P}}^2} \\ &\equiv -\pi \sum \omega^{-1}(a) T(a). \end{aligned}$$

Regarding $\mathbb{F}_q$ as $\mathbb{Z}[\zeta_{q-1}] \pmod{\mathfrak{p}}$, we have

$$T(a) = a + a^p + \cdots + a^{p^{f-1}} \pmod{\mathfrak{p}}$$

(since $a \mapsto a^p$ generates the Galois group $\pmod{\mathfrak{p}}$), and

$$\sum \omega^{-1}(a) T(a) \equiv \sum_{\substack{a \not\equiv 0 \\ a \pmod{\mathfrak{p}}}} a^{-1} (a + a^p + \cdots + a^{p^{f-1}}) \pmod{\mathfrak{p}}.$$

If $0 < b < f$, then $\sum_{a \not\equiv 0} a^{p^b-1} \equiv 0 \pmod{\mathfrak{p}}$, so the sum reduces to $\sum_{a \not\equiv 0} 1 = q-1 \equiv -1$. Therefore

$$g(\omega^{-1}) \equiv \pi \pmod{\tilde{\mathcal{P}}^2},$$

hence $s(1) = v_{\tilde{\mathcal{P}}}(\pi) = 1$ (since $\mathbb{Q}(\zeta_{q-1}, \zeta_p)/\mathbb{Q}(\zeta_p)$ is unramified at p). This completes the proof. $\square$

Proposition 6.13. Let $0 \leq \alpha < q-1$ and let $\alpha = a_0 + a_1 p + \cdots + a_{f-1} p^{f-1}$, $0 \leq a_i \leq p-1$, be the standard p -adic expansion of α . Then

$$s(\alpha) = a_0 + a_1 + \cdots + a_{f-1}.$$

Proof. From Lemma 6.11(a), (b), (c) and Lemma 6.12 we immediately have $s(\alpha) = \alpha$ for $0 \leq \alpha \leq p-2$. If $q = p$, we are done. Otherwise, $s(p-1) > 0$

and we similarly obtain $s(p-1) = p-1$. Lemma 6.11(b) and (d) imply that $s(\alpha) \leq a_0 + \cdots + a_{f-1}$. When α runs through the integers from 0 to $q-1$, inclusive, each coefficient of the p -adic expansion takes on each of the values from 0 to $p-1$ exactly p^{f-1} times, so

$$\sum_{\alpha=0}^{q-1} (a_0 + \cdots + a_{f-1}) = \frac{p(p-1)}{2}(f)p^{f-1} = \frac{p-1}{2}fq.$$

If we omit $\alpha = q-1 = (p-1) + \cdots + (p-1)p^{f-1}$, we obtain

$$\sum_{\alpha=0}^{q-2} (a_0 + \cdots + a_{f-1}) = \frac{p-1}{2}fq - (p-1)f = \sum_{\alpha=0}^{q-2} s(\alpha),$$

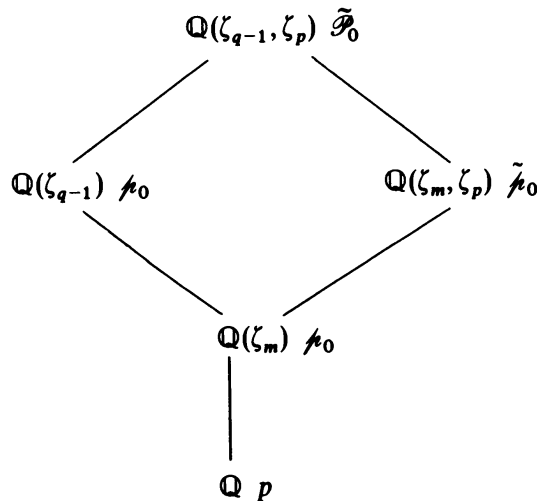
by Lemma 6.11(e). The result follows. $\square$

Remark. Let $\pi = \zeta_p - 1$ and $0 \leq \alpha < q-1$, as above. Then

$$g(\omega^{-\alpha}) \equiv \frac{\pi^{a_0 + \cdots + a_{f-1}}}{(a_0!) \cdots (a_{f-1}!)} \pmod{\tilde{\mathcal{P}}^{a_0 + \cdots + a_{f-1} + 1}}.$$

(see Lang [4], [5]). In Lemma 6.12 we verified a special case of this formula. The general argument follows a similar line, but involves a rather delicate analysis of binomial coefficients.

Now fix a positive integer m . Let p be a prime, $(p, m) = 1$, and let f be the order of $p \pmod{m}$, so m divides $p^f - 1 = q - 1$. Fix a prime $\mathfrak{p}_0$ of $\mathbb{Q}(\zeta_m)$ lying above p ; let $\tilde{\mathfrak{p}}_0$ be the prime of $\mathbb{Q}(\zeta_m, \zeta_p)$ above $\mathfrak{p}_0$, so $\tilde{\mathfrak{p}}_0^{p-1} = \mathfrak{p}_0$; let $\mathcal{P}_0$ be a prime of $\mathbb{Q}(\zeta_{q-1})$ lying above $\mathfrak{p}_0$; and let $\tilde{\mathcal{P}}_0$ be the prime of $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ lying above $\mathcal{P}_0$ (and $\tilde{\mathfrak{p}}_0$). Let $\omega = \omega_{\mathcal{P}_0}$ be as above and let $\chi = \omega^{-d}$, where $d = (q-1)/m$. Then $\chi^m = 1$, so $g(\chi) \in \mathbb{Q}(\zeta_m, \zeta_p)$. Since $g(\chi)\overline{g(\chi)} = q = p^f$, the factorization of $g(\chi)$ involves only primes of $\mathbb{Q}(\zeta_m, \zeta_p)$ above p , that is, the conjugates over $\mathbb{Q}$ of $\tilde{\mathfrak{p}}_0$. Let $(a, m) = 1$ and let $\sigma_a \in \text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ be the corresponding element of the Galois group. For each such a , fix an extension of σ_a to $\mathbb{Q}(\zeta_{q-1}, \zeta_p)$ such that $\zeta_p^{\sigma_a} = \zeta_p$.



The decomposition group for p in $(\mathbb{Z}/m\mathbb{Z})^\times$ is generated by $p \pmod{m}$ (see the discussion after Theorem 2.13). Let R denote a set of representatives for $(\mathbb{Z}/m\mathbb{Z})^\times$ modulo this decomposition group. Then $\{\tilde{\rho}_0^{\sigma_a^{-1}} | a \in R\}$ is the set of conjugates of $\tilde{\rho}_0$. Since $\tilde{\rho}_0$ is the unique prime above ρ_0 , all conjugates of $\tilde{\rho}_0$ have the form $\tilde{\rho}_0^{\sigma_a^{-1}}$. Let $\tilde{\rho} = \tilde{\rho}_0^{\sigma_a^{-1}}$ be one of them. Then

$$v_{\tilde{\rho}}(g(\chi)) = v_{\tilde{\rho}_0}(g(\chi)^{\sigma_a}) = v_{\tilde{\rho}_0}(g(\chi^a)) = v_{\tilde{\rho}_0}(g(\chi^a)) = s(ad)$$

($v_{\tilde{\rho}_0} = v_{\tilde{\rho}}$ since $\tilde{\mathcal{P}}_0/\tilde{\rho}_0$ is unramified). Therefore

$$(g(\chi)) = \tilde{\rho}^{\sum_{a \in R} s(ad)\sigma_a^{-1}}.$$

Lemma 6.14. *Let $0 \leq h < q - 1$. Then*

$$s(h) = (p - 1) \sum_{i=0}^{f-1} \left\{ \frac{p^i h}{q - 1} \right\}.$$

Proof. Let $h = a_0 + a_1 p + \cdots + a_{f-1} p^{f-1}$. Then

$$p^i h \equiv a_0 p^i + a_1 p^{i+1} + \cdots + a_{f-1} p^{i-1} \pmod{q - 1}.$$

It follows that

$$\left\{ \frac{p^i h}{q - 1} \right\} = \frac{1}{q - 1} (a_0 p^i + \cdots + a_{f-1} p^{i-1}).$$

Summing over i , we obtain the result. □

We now have $s(ad) = (p - 1) \sum_{i=0}^{f-1} \{p^i a/m\}$, so

$$\sum_R s(ad) \sigma_a^{-1} = (p - 1) \sum_{i=0}^{f-1} \sum_R \left\{ \frac{p^i a}{m} \right\} \sigma_a^{-1}.$$

Since $\tilde{\rho}_0^{p-1} = \rho_0$ and since $\sigma_{p^i}(\rho_0) = \rho_0$ (definition of decomposition group),

$$(g(\chi)^m) = \rho_0^m \sum \sum \{p^i a/m\} \sigma_{a p^i}^{-1} = \rho_0^{m\theta},$$

where $\theta = \sum_{b=1, (b,m)=1}^m \{b/m\} \sigma_b^{-1}$ is the Stickelberger element (we raise to the m th power to avoid denominators).

We now have a partial result: If ρ_0 is a prime of $\mathbb{Q}(\zeta_m)$ with $\rho_0 \nmid m$, then $\rho_0^{m\theta}$ is principal in $\mathbb{Q}(\zeta_m, \zeta_p)$. The main problem is now to get down to $\mathbb{Q}(\zeta_m)$, then to M .

Suppose that A is an ideal of $M \subseteq \mathbb{Q}(\zeta_m)$ with $(A, m) = 1$. Let $A = \prod \rho_i$ be its factorization into (not necessarily distinct) prime ideals in $\mathbb{Q}(\zeta_m)$. Then

$$A^{m\theta} = \left(\prod g(\chi_{\rho_i})^m \right),$$

where we write χ_{ρ_i} to indicate that χ depends on ρ_i . Suppose $\beta \in \mathbb{Z}[G]$ ($G = \text{Gal}(M/\mathbb{Q})$) and $\beta\theta \in \mathbb{Z}[G]$. Extending the elements of G , we may regard $\beta\theta$ as an element of $\mathbb{Z}[\text{Gal}(\mathbb{Q}(\zeta_{mp})/\mathbb{Q})]$. Then

$$A^{m\beta\theta} = (\gamma^{\beta m}), \quad \text{where } \gamma = \prod g(\chi_{\rho_i}) \in \mathbb{Q}(\zeta_{pm}),$$

where $P = \prod p_i$ is the product of the rational primes divisible by the $\mathfrak{p}_i$'s. Since $\gamma^{m\beta} \in \mathbb{Q}(\zeta_m)$ by Lemma 6.4, and it is the m th power of an ideal of $\mathbb{Q}(\zeta_m)$, namely $A^{\beta\theta}$, it follows that the extension $\mathbb{Q}(\zeta_m, \gamma^\beta)/\mathbb{Q}(\zeta_m)$ can be ramified only at primes dividing m (*proof*: locally, $A^{\beta\theta}$ is principal, so we are adjoining the m th root of a local unit). But

$$\mathbb{Q}(\zeta_m) \subseteq \mathbb{Q}(\zeta_m, \gamma^\beta) \subseteq \mathbb{Q}(\zeta_m, \zeta_P).$$

Therefore, ramification can occur only at p_i 's. Since $(P, m) = 1$, the extension must be unramified.

Lemma 6.15. *If $\mathbb{Q}(\zeta_m) \subseteq K \subseteq \mathbb{Q}(\zeta_n)$ and $K/\mathbb{Q}(\zeta_m)$ is unramified at all primes, then $K = \mathbb{Q}(\zeta_m)$.*

Proof. Suppose $K \neq \mathbb{Q}(\zeta_m)$. Then there is a character χ for K of conductor not dividing m . By Theorem 3.5, $K/\mathbb{Q}(\zeta_m)$ must be ramified at some prime. Contradiction. For another proof, see Lemma 15.48. $\square$

We find that $\gamma^\beta \in \mathbb{Q}(\zeta_m)$. Therefore $A^{\beta\theta} = (\gamma^\beta)$ is principal as an ideal of $\mathbb{Q}(\zeta_m)$. But this does not necessarily mean that it is principal as an ideal of M . So we show that $\gamma^\beta \in M$, which suffices, since if two ideals of M are equal in $\mathbb{Q}(\zeta_m)$ they must have been equal originally because of unique factorization.

Let $\mathcal{P}$ be a prime of $\mathbb{Q}(\zeta_{q-1})$ lying over one of the prime factors $\mathfrak{p}_i$ of A . Then $\chi_{\mathfrak{p}_i}$ a priori depends on the choice of $\mathcal{P}$, so we temporarily let $\chi_{\mathfrak{p}_i} = \chi_{\mathcal{P}}$. Let $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_{q-1})/M)$. Then

$$\sigma: \mathbb{Z}[\zeta_{q-1}] \bmod \mathcal{P} \xrightarrow{\sim} \mathbb{Z}[\zeta_{q-1}] \bmod \mathcal{P}^\sigma$$

and correspondingly if $\chi_{\mathcal{P}}(a) = \zeta$ then $\chi_{\mathcal{P}^\sigma}(a) = \zeta^\sigma$. Therefore $\chi_{\mathcal{P}}^\sigma = \chi_{\mathcal{P}^\sigma}$. But $\chi_{\mathcal{P}}^m = 1$, so $\chi_{\mathcal{P}^\sigma} = \chi_{\mathcal{P}}$ for $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_{q-1})/\mathbb{Q}(\zeta_m))$. Therefore $\chi_{\mathcal{P}}$ depends only on $\mathfrak{p}_i$, so we may return to the notation $\chi_{\mathfrak{p}_i}$. The above reasoning shows that $\chi_{\mathfrak{p}_i}^\sigma = \chi_{\mathfrak{p}_i^\sigma}$ for $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)/M)$. If we extend σ by letting $\sigma(\zeta_p) = \zeta_p$, then $g(\chi_{\mathfrak{p}_i})^\sigma = g(\chi_{\mathfrak{p}_i}^\sigma) = g(\chi_{\mathfrak{p}_i^\sigma})$.

Since $A^\sigma = A$ for $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_m)/M)$, σ permutes the $\mathfrak{p}_i$'s. Therefore

$$\gamma^{\beta\sigma} = \prod g(\chi_{\mathfrak{p}_i})^{\beta\sigma} = \prod g(\chi_{\mathfrak{p}_i^\sigma})^\beta = \gamma^\beta.$$

But we already have $\gamma^\beta \in \mathbb{Q}(\zeta_m)$; hence $\gamma^\beta \in M$. So $A^{\beta\theta}$ is principal in M .

Finally, if A is an arbitrary ideal of M , we may write $A = (a)A_1$, with $a \in M$ and $(A_1, m) = 1$. Then

$$A^{\beta\theta} = (a^{\beta\theta})A_1^{\beta\theta},$$

which is principal. This completes the proof of Stickelberger's theorem. $\square$

For an easier proof of Stickelberger's theorem in the case of a full cyclotomic field $\mathbb{Q}(\zeta_m)$, see Section 15.1.